

Offline preparation

Raw CICIoT2023 CSVs
46 features, 34 labels
~46.7M rows

46 feat.

Feature selection
drop zero/low-variance
drop high-corr. (> 0.95)

29 feat.

Kill-chain projection
 $\pi_{\text{kc}} : \{1, \dots, 34\} \rightarrow \{0, \dots, 4\}$

5 stages

Per-stage distributions
 $p_{\text{data}}(x \mid s)$



Episode trajectory τ
 $(o_1, a_1, r_1, \dots, o_T, a_T, r_T)$

Window stack ϕ
 $o_t \in \mathbb{R}^{290}$

Observation kernel Z
aliasing at rate α

Online environment